

TREVOR CROUSE

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EDUCATION

CORNELL UNIVERSITY, Duffield College of Engineering, Ithaca, New York

Bachelor of Science Degree in Mechanical Engineering: Expected May 2028

- **Relevant Coursework:** Dynamics; Statics and Mechanics of Solids; Thermodynamics, Mechanical Design; Electromagnetism; Waves and Quantum Physics; Differential Equations; Linear Algebra
- **Campus Involvement:** American Society of Mechanical Engineers, Aerodynamic Research, Makers Club

SPECIALIZED SKILLS AND CERTIFICATIONS

- ANSYS Fluent; CFD; SolidWorks CAD; Fusion 360; MathWorks MATLAB; Python; Aerodynamic Analysis; Mechanical Design; Machine Shop; Microsoft Word/Excel/PowerPoint; Technical Documentation
- AutoCAD Basic Training (Jan. 2026 – Udemy); Airplane Design and Aerodynamic Basic (Oct. 2024 – Udemy)
- Automobile Fundamentals and Advanced (Oct. 2024 – ValueKnow)

RELEVANT EXPERIENCE

Design, Build, Fly Project Team; Cornell University; Ithaca, New York

Aerodynamics and Controls Sub-team Lead

November 2024 – Present

- Directed a 14-member aerodynamic team to create all wing and tail components of a high-performance RC aircraft. Utilized SolidWorks CAD for design and ANSYS Fluent for aerodynamic performance
- Collaborated with cross-functional sub-team to design and integrate subsystems, using ANSYS Mechanical FEA to perform stress analysis on components and to create simulations of ground competition missions
- Led technical documentation on a 58-page competition report for the annual *AIAA Design, Build, Fly* competition.
- Competed in Wichita, Kansas against different U.S. and international universities

Zeta Psi Fraternity; Cornell University; Ithaca, New York

Executive Board Position for Event Planning and Risk Management

April 2025 – Present

- Managed / Allocated a \$30,000 budget for events, including catering, venue selection, and vendor coordination
- Developed / Implemented risk management strategies, including transportation and safety protocols for over 90 people
- Coordinated logistics with internal committees / external partners to ensure smooth execution of large-scale events

PROJECTS

Aircraft Wing Design and Manufacturing for Design, Build Project; Ithaca, New York

August 2025 – Present

- Optimized airfoil and wing shape around AIAA constraints using XFLR5 and a self-made MATLAB sizing script
- Designed a manufacturable full-scale wing in SolidWorks creating useful assemblies to test aileron deflections and to use in ANSYS CFD to perform lift and drag analysis
- Manufactured the wing, iterating five different versions reducing mass and decreasing induced drag using hardwood, balsa, and 3D designed integration pieces

BuzzKill: Mechanical Design & Control Project for an Invasive Insect; Ithaca, New York

January 2026 – Present

- Designed a mechanical solution to control spotted lanternfly populations in vineyards, commissioned by E. & J. Gallo Winery and National Grape
- Tested various solutions, deciding on a jaw bucket design attached to an extendable rod using high mechanical advantage to reduce labor intensity, creating a scalable and cost-efficient solution
- Selected high strength, weather resistant materials and applied DFM principles to optimize the jaw bucket for field durability and low-cost manufacturing

Wing Endplate Aerodynamic Research; Ithaca, New York

January 2026 – February 2026

- Conducted research on wing endplates efficiency and contribution to lift at low Reynolds Numbers, using professor insight to design endplate shapes in SolidWorks and using ANSYS CFD to perform lift-drag analysis
- Wrote a report on the effectiveness of wing endplates, balancing their effectiveness while comparing their benefits with weight and manufacturing constraints